

TECHNICAL ARTICLES

PATHWAYS OF DYE TRACER MOVEMENT THROUGH STRUCTURED SOILS ON A MACROSCOPIC SCALE

M. M. Nobles, L. P. Wilding, and K. J. McInnes

Structural features, biopores, and slickensides have been reported to be important features impacting bypass flow of pollutants to shallow groundwater aquifers. The objectives of this research were to identify the pathways of water and solute transport at a macroscopic scale and to identify visible characteristics of solute flow in two Vertisols and one Alfisol. Relative involvement of structural features in solute transport, such as ped interfaces, root channels, and slickenside planes, were studied by mapping dye flow paths. Roots and interpedal pores served as major pathways of transport in all three soils. Slickenside fissures in Vertisols served as minor pathways of solute transport. Solute flow along slickenside surfaces was observed primarily when the dye solution was delivered to the surface of the slickenside via the adjacent root channel. The differences in observed distribution of dye pathways were related to the abundance of different types of structural features, their apparent continuity, interconnectivity, and macropore access to the infiltrating dye front. Lateral transport was identified as one of the major components of the flow, even though the sites were located on nearly-level to gently sloping geomorphic surfaces. (Soil Science 2004;169:229-242)

Key words: Preferential flow, Vertisols, tracer movement, slickenside, macropores.

UNDERSTANDING the mechanisms of solute transport in soils is necessary to predict the fate of chemicals applied to soil. Studying transport in structured soils, Vertisols and vertic intergrades in particular, is of importance because of the essential role of macropores. Structured soils are prone to producing bypass flow, with consequent deep solute penetration (Flury and Flühler, 1994), making application of chemicals to these soils a potential hazard for groundwater contamination.

The importance of preferential flow in cracking clayey soils has long been recognized (Lawes et al., 1882; Bouma and Dekker, 1978; Steenhuis and Parlange, 1991). Cracking clayey soils are characterized by low permeability of the soil matrix combined with the presence of continuous vertical desiccation cracks and conduits created by roots and soil fauna. According to Bouma and Dekker (1978), the number, continuity, and surface characteristics of larger vertical pores are major factors influencing the patterns of solute infiltration into dry clay soils. Rainwater or irrigation water may move rapidly along these channels, bypassing the slowly penetrated surrounding matrix. In a case of swelling clayey soil in Texas (Houston Black series), a tracer was detected as deep as 90 cm from the soil surface after only one and a half days of ponding as a

Texas A&M University, Dept. of Soil and Crop Sciences, College Station, TX 77843-2474. Dr. Nobles is corresponding author. E-mail: m-morgun@tamu.edu

Received July 15, 2003; accepted Jan. 7, 2004.

DOI: 10.1097/01.ss.0000126838.81093.20

result of the presence of natural structural interfaces (Kissel et al., 1973).

The presence of macropores in cracking clay soils, combined with excessive fertilizer and manure application, can result in the fast preferential transport of chemicals such as pesticides into shallow water table depths, increasing the risk of groundwater contamination. It has been established that nutrients such as P can be transported rapidly along these macropores to the drainage systems (Stamm et al., 1998), creating a potential for eutrophication of rivers and lakes. Preferential flow was found to be an important component of N transport in soil as well (Kanwar and Bakhsh, 2001). Preferential transport is also important from a nutrient loss perspective. When macropores in soils are major conduits of solutes, surfaces of these preferential pathways soon reach their capacity to retain P, N, and toxic chemicals, and, thus, the buffering capacity of the system is filled. This allows for vertical and lateral transport at much faster rates and deeper depths than would ordinarily occur.

Identification of the nature of preferential flow pathways in a given soil is also essential for the prediction of contaminant behavior. Gish and Jury (1983) and Gish et al. (1998) emphasized the importance of root channel macropores in decreasing the transit time of solutes to underlying groundwater. The study by Mitchell et al. (1995) showed that macropores formed by large-diameter straight taproots, such as those of alfalfa, have greater potential for solute transport than macropores formed by fine shallow roots or desiccation cracks. Preferential transport of chemicals along the root channels also increases the access of plant roots to a contaminating agent, which, in turn, could increase the possibility of uptake or phytoremediation (Clothier and Green, 1997, Cunningham et al., 1996). Likewise, the transport of solutes through pathways formed by dead roots could be significant in contaminant absorption by organic material. Turner and Steele (1988) showed that organic-rich linings and fillings in root-derived soil macropores exhibited higher sorptivities for cadmium and manganese than did inorganic linings or homogenized soil.

The preferential movement of solutes along the surfaces of peds characterized by the presence of clay coatings, such as those in Alfisols, could increase the possibility of contaminant sorption by clays (Beck and Jones, 1996). This aspect could also be pertinent to oxidation/reduction reactions adjacent to these pathways, which could influence the change in oxidation state, solubility,

and mobility of contaminants. Slickenside fissures are also considered to be one of the major pathways of water in Vertisols (Wilding and Tessier, 1988). Slickenside surfaces in Vertisols and Vertic intergrades are generally thought to act as an important pathway of solute movement by providing inflow from microhighs to microlows.

Although all of the above-mentioned pathways have been identified as contributing to preferential flow, relative distribution of water flux among structural features remains debatable. The study by Lin et al. (1996) found that slickenside fissures in Texas Vertisols contributed 30% of the flow paths, root channels 28%, and vertical fissures 25%; the remaining 17% were associated with unidentified soil morphological features.

The objectives of this research were to verify patterns and mechanisms of water and solute movement on macroscopic scale and to develop an understanding of water flux distribution among the groups of structural features in three soils with different degrees of structural development.

MATERIALS AND METHODS

Study Area and Soils

The study was conducted on two Vertisols (Ships and Burleson series) and one Alfisol (Silawa variant), all of which were located in southeast Texas. Conducting the research on these particular soils provided the opportunity to study water transport in soils that contain a range of structural properties. In addition, the sites of Ships clay and Burleson clay were chosen because of extensive previous research work that served as a background for further study of water transport. The moisture and temperature regimes of the area are classified as ustic and thermic, respectively (Mowery et al., 1958), and physical and chemical properties of each soil were also characterized. For the Burleson clay series, the existing data were used (Lin et al., 1996). Analyses included organic matter content (method code 6A), pH (method code 8C), cation exchange capacity (method code 5A), and calcium carbonate equivalent (method code 6E) (Soil Survey Staff, 1996). Selected properties of the three soils used in this study are presented in Table 1.

The site of the Ships series was located on a nearly level floodplain of the Brazos River. The parent material consisted of calcareous clayey sediments of Pleistocene age deposited by floodwaters of the Brazos River (Mowery et al., 1958). The sampling site was left under native grassland vegetation. The Ships series was classified as fine,

TABLE 1
The soils investigated in this study and their basic properties

Horizon	Depth cm	Texture	Total clay %	CEC cmol/kg	Organic carbon %	CEC/clay cmol/kg	pH (1:1 H ₂ O)	CCE† %
<u>Ships clay (fine, mixed, superactive, thermic Chromic Haplusterts)</u>								
Ap1	0-10	Silty Clay Loam	32.4	26.0	1.09	80	8.0	12.7
Ap2	10-25	Silty Clay Loam	37.7	28.2	0.96	75	8.0	13.7
Bssk1	25-60	Silty Clay Loam	38.8	27.2	0.67	70	8.0	12.2
Bssk2	60-80	Silty Clay	41.4	29.6	0.70	71	8.0	11.8
Bssk3	80-120	Clay	61.8	41.1	0.86	67	7.9	10.3
<u>Burleson clay (fine, smectitic, thermic Udic Haplusterts)‡</u>								
<u>Microhigh</u>								
A	0-20	Clay	50.6	45.0	1.42	89	7.6	11.3
Bssk1	20-63	Clay	53.9	43.0	0.45	80	7.8	12.9
Bssk2	62-147	Clay	52.1	42.5	0.14	82	7.9	18.9
<u>Microlow</u>								
A	0-20	Clay	60.9	57.5	1.98	94	7.3	0.4
BA	20-50	Clay	63.2	51.9	1.30	82	7.4	1.1
Bssk	50-166	Clay	61.3	48.7	0.95	79	7.6	3.2
<u>Silawa variant (clayey, smectitic, thermic Ultic Haplustalfs)</u>								
<u>Microhigh</u>								
Ap	0-5	Loamy Fine Sand	3.8	5.7	1.03	150	4.2	0.0
E	5-30	Fine Sandy Loam	9.6	7.3	0.46	76	5.2	0.0
Bt/E	30-55	Sandy Clay	37.5	30.7	0.31	82	4.7	0.0
Bt1	55-80	Clay	41.6	37.4	0.20	90	4.4	0.0
Bt2	80-95	Clay	57.8	50.6	0.13	86	6.6	0.0

†CCE - Calcium Carbonate Equivalent.

‡Lin et al. (1996).

mixed, superactive, thermic Chromic Haplusterts. The Bssk horizons had abundant slickensides at a depth of 25 to 120 cm. The Ships clay also had many biological features, such as roots and worm channels. The root channel system in Ships clay consisted of fine and very fine live roots as well as root channels formed around roots dead at the time of the study.

The site of the Burleson series was located on a terrace of the Brazos River. The parent material consisted of alkaline, clayey, alluvial sediments of Pleistocene age. The site had recently been cultivated with a subsurface sweep, but the precise location of the experimental site was near the border of the field where there was minimal disturbance. The Burleson series was characterized as fine, smectitic, thermic Udic Haplusterts. It had abundant large slickenside planes at depths from 50 to 166 cm. Some slickenside planes achieved the dimensions of 0.8 to 1 m in length and lateral extent. The landscape of the Burleson series was characterized by normal gilgai relief represented by distinct microhighs and microlows less than 4 m apart and with amplitude of about 20 cm. The microhighs and microlows of the Burleson series had different vegetative covers. Juniper leaf, plantain, white tridens, and heath aster were dominant on the microhighs. Broomweed, buffalograss, and sorrel were found most frequently in the microlows. The vegetation in Burleson clay microhighs had medium and coarse (2–5 mm) roots that formed continuous channels to a depth of approximately 1 m, whereas the vegetation in microlows had relatively shallow (approximately 30 cm) root systems consisting of fine roots.

The Silawa variant was located on a gently sloping terrace in Brazos County. The parent material of the Silawa variant consisted of loamy sediments of Pleistocene age. The site was cultivated under cotton from approximately 1870 to 1920. It was then left in fallow until 1976 when it was converted to lawn under Bermuda grassland vegetation. The Silawa variant was classified as clayey, smectitic, thermic Ultic Haplustalfs. The profile had a loamy fine sand surface Ap horizon and a fine sandy loam E horizon over a well developed clay argillic horizon containing common clay cutans on ped faces. The Bt/E horizon was characterized by tongues and pockets of coarser material infilling the cracks in the sandy clay loam material of the Bt1 horizon. The Silawa variant contained common to many roots (1 to 5 roots / cm²). In the subsoil horizons, continuous medium live roots formed common root

channels. Few very coarse (>10 mm) living pine roots were observed.

Tracer Characteristics

Brilliant Blue FCF (C₃₇H₃₄N₂Na₂O₉-S₃) was used as a dye tracer to identify pathways of solute transport visually. It is a nontoxic, water-soluble organic dye of dark blue color, highly visible in most soils. A study by Flury and Flühler (1995) has suggested that in soil, Brilliant Blue FCF is either neutral or dissociates to a mono- or bivalent anion, depending on pH (pK_a = 5.83 and 6.58).

Dye-Tracing Study of Water Flow

On each site, four PVC cylinders, each with an inner diameter of 35 cm, were driven manually into the soil to a depth of 15 cm. The diameter of the cylinder was chosen to be approximately 10 times larger than the representative elemental units in the soils studied (2–3 cm). On the Burleson series, two cylinders were positioned on microhighs and two on microlows. However, upon excavation, one of the duplicates of the Burleson clay microhigh site was lost. Therefore, the Burleson series in this study is represented by one microhigh and two microlow observations. Fifty-three liters of a tracing solution of Brilliant Blue FCF at a concentration of 30 gL⁻¹ and CaBr at 60 gL⁻¹ were applied to each of the cylinders. The CaBr solution was used in a later study to estimate the distribution of Br on the submicroscopic level by means of electron microprobe analysis (Nobles et al., 2003). Fifty-three liters constituted approximately two pore volumes of a soil column 35 cm in diameter to a depth of 1.5 m with a moisture content of 25%. Thus, the volume of solution was calculated so that the solution would fill the pores completely to the depth of 150 cm, creating saturated flow. The dye solution was applied to the cylinders manually in multiple increments, with a head of approximately 5 cm, and allowed to infiltrate before the next increment was added. The solution was added continuously to allow macropores to be filled completely throughout the solute application process.

The soil was excavated within 10 to 12 h after the last portion of the dye solution had completely infiltrated the soil, thus allowing the macropores to empty. Excavation was done with two trenches, each of which included two infiltration sites. The trenches were excavated through the center of the infiltration zone. Each trench was approximately 6 m long and 2 m deep. The exposed vertical face was then cleaned

with the shovel and carefully picked with the knife to prevent the smearing of the dye. Exposed dye patterns were traced onto transparent plastic sheets (Petersen et al., 1997) and color photos of dye patterns were taken.

Morphological features involved in the transport of dye, such as ped interfaces, biopores, and slickenside planes, were identified in the field and labeled on plastic sheets. The solute-transporting features were grouped into the following classes: root channels, ped interfaces, biological features such as worm channels and termitarium, slickensides, undifferentiated features, and nonidentified features. The term "undifferentiated features" was used to indicate a combination of root channels and ped interfaces, usually located in the upper horizon of Ships clay and Burleson clay, that were stained completely along ped surfaces so that no differentiation of individual features could be made. In the case of the Silawa variant, this term was used to identify completely stained root channels and packing voids located in the Ap and E horizons. The term "unidentified" was used to indicate features that could not be readily classified.

The images were digitized, converted to black and white binary images, and analyzed with Adobe Photoshop 6.0. The total area of the dye-stained section was calculated, and the percentage of dye-stained area contributed by each type of feature was estimated as a fraction of the total area for every soil horizon and for the entire profile.

The lateral movement of the dye was estimated by calculating the percentage of the total stained area contributed by dye-stained features outside the zone located directly below the infiltration cylinder.

To determine the moisture condition of the soil at the time the dye-staining treatments were conducted, soil samples were collected approximately 2 m outside the area of the infiltration cells immediately after the excavation. Samples were stored in airtight metal containers and transported to the laboratory where gravimetric moisture contents were determined (Soil Survey Staff, 1996). Water retention was determined by the pressure plate extraction method at 0.03, 0.1, and 1.5 MPa (Soil Survey Staff, 1996). The values of water tension for field moisture contents were estimated on the basis of the soil water retention curve.

RESULTS AND DISCUSSION

Field Condition of Studied Soils

Field moisture contents of these soils and the corresponding soil moisture tension values are shown in Table 2. It was estimated that in Burleson clay, the tension values varied from 0.9 MPa (A horizon) to 1.4 MPa (Bssk horizon). Tension values in Ships clay varied from 0.2 MPa (Bssk2 horizon) to 1.4 MPa (Ap1 horizon). The absence of major cracks in both Ships clay and Burleson clay was observed visually at the time of

TABLE 2
Gravimetric moisture content and soil moisture tension values of three studied soils

Horizon	Depth, cm	Moisture content, g/g	Soil moisture tension, MPa	Moisture content at		
				0.03 MPa, g/g	0.1 MPa, g/g	1.5 MPa, g/g
<u>Ships clay</u>						
Ap1	0–10	0.17	1.4	0.30	0.25	0.16
Ap2	10–25	0.22	0.6	0.30	0.24	0.19
Bssk1	25–60	0.24	0.8	0.32	0.28	0.20
Bssk2	60–80	0.26	0.2	0.30	0.25	0.20
Bssk3	80–100	0.28	0.7	0.36	0.32	0.22
<u>Burleson clay</u>						
A	0–20	0.26	0.9	0.35	0.31	0.22
BA	20–50	0.24	1.3	0.34	0.31	0.23
Bssk	50–166	0.25	1.4	0.41	0.35	0.24
<u>Silawa variant</u>						
Ap	0–5	0.18	<0.03	0.11	0.10	0.04
E	5–30	0.16	<0.03	0.08	0.08	0.08
Bt/E	30–55	0.17	<0.03	0.15	0.13	0.10
Bt1	55–80	0.19	1.5	0.28	0.26	0.19
Bt2	80–95	0.17	>1.5	0.27	0.24	0.18

the experiment. The absence of open desiccation cracks has a direct influence on the type of solute pathways. Kutílek (1996) suggested that when cracking clay soils are wetted, the majority of cracks are reduced in size and transformed into mesopores of dimensions approaching those of the interpedal pores. The absence of cracks as readily available macropore pathways for solute transport could cause the relative increase in the involvement of macropores formed by root channels. The soil moisture conditions of the fine sandy loam surface horizons of Silawa variant were characterized by tensions lower than 0.03 MPa, whereas in the clay subsoil horizons, the values of soil moisture tension were at or greater than 1.5 MPa.

Dye Movement in Pedons of Ships Clay

The calculations of stained areas contributed by features involved in solute transport showed that in surface horizons of four Ships clay sites, the majority of the stained area (86%) was attributed to undifferentiated features (Table 3). Among other features transmitting dye solution

in Ships clay surface horizons were ped interfaces (14%) and root channels (3%) (Fig. 1).

In subsurface horizons of Ships clay sites, stained areas were associated primarily with ped interfaces (18%) and root channels (37%). Visual observation of stained root channels in the field has shown that solute flow was conducted along living roots as well as along vacant root channels formed by previous roots. However, the differentiation between live and dead roots in stained root channels was hard to determine in the field, presenting a high probability for error.

Less than 1% of the stained surfaces in the subsoil horizons of site 4 were attributed to the flow along the slickenside planes. Limited flow along slickenside faces was also observed visually in three other sites of Ships clay during sample collection. These findings are different from those of Lin et al. (1996), who found that fissures in Ships clay might constitute 30% of flow paths in slickenside. However, the results shown in the study by Lin et al. (1996) were obtained when an infiltrometer was placed directly on the surface of the Bssk horizon, creating connectivity between

TABLE 3

Percentages (as fraction of total stained area) of dye-stained features involved in solute transport at four sites on Ships clay

Horizon	Ped interfaces	Root channels	Undifferentiated features [†]	Unidentified features	Slickenside planes	Worm channels
Ap (0–25 cm)						
Site 1	8	5	85	2	0	<1
Site 2	<1	3	97	<1	0	0
Site 3	30	<1	68	<1	0	0
Site 4	3	2	95	0	0	0
Mean	14	3	86	1		
SD	12	1	11	1		
CV	86	37	13	100		
Bssk (25–120 cm)						
Site 1	27	26	0	47	0	0
Site 2	15	46	0	39	0	0
Site 3	23	34	13	30	0	0
Site 4	8	42	25	25	<1	0
Mean	18	37		35		
SD	7	8		8		
CV	40	21		24		
Depth of dye flow, cm						
Site 1		62				
Site 2		76				
Site 3		100				
Site 4		80				
Mean		80				
SD		16				
CV		20				

[†]Undifferentiated root channels and ped interfaces.

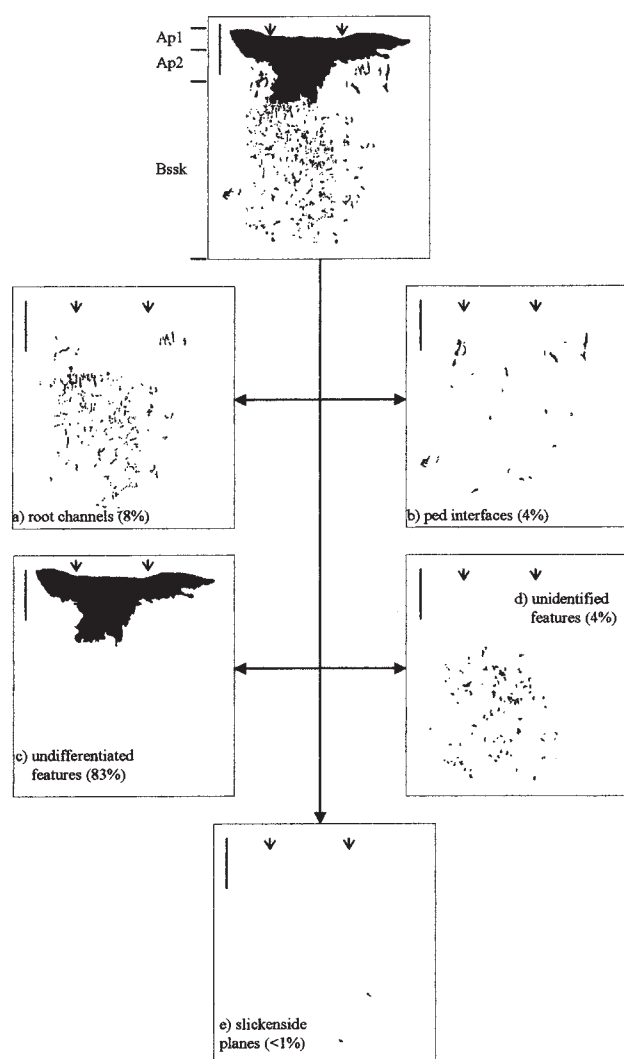


Fig. 1. Two-dimensional dye flow pattern in site 4 of Ships clay exposed along the central vertical line of intake zone (indicated by arrows). Overall dye-stained areas are separated into areas associated with (a) root channels, (b) ped interfaces, (c) undifferentiated features, (d) unidentified features and (e) slickenside planes. The vertical bar indicates 20 cm.

the infiltrometer and the macropores and allowing for the direct access of the dye solution to the slickenside surfaces, thus possibly resulting in the increased involvement of slickenside fissures in solute transport.

Involvement of slickenside surfaces in solute transport was detected primarily when the dye solution was transported to the slickenside by the adjacent root channels located in the immediate vicinity of the slickenside fissure. The dye-stained areas in these cases were identified as the result of the dye transport along the slickenside surface. The solution often moved along very fine cylindrical root channels, penetrating the surface of the slickenside. In such cases, the contact of the root channel with the slickenside fissure would cause the dye solution to advance along the sur-

face of the slickenside and stain a small (approximately 3–4 mm) area on the surface of the slickenside immediately adjacent to the root channel (Fig. 2). The primary transport of the solution would then continue along the root channel. In some instances, stained root channels were positioned between the abutting surfaces of slickenside fissures (Fig. 2). The spatial distribution of dye on the surface of slickenside in these cases was often limited to the vicinity of the root channel, without the movement of the dye solution further along the slickenside surface.

Statistical analysis of the data collected in four sites of Ships clay demonstrated high variability in size of stained areas contributed by dye-conducting structural features. Coefficients of variation ranged from 13% (areas contributed by un-

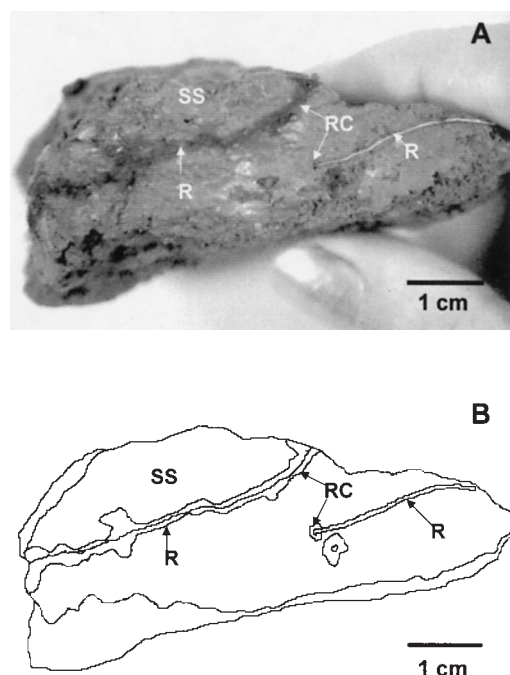


Fig. 2. (A) Slickenside plane (SS) in the Bssk horizon of Ships clay stained by Brilliant Blue FCF dye that has been transported to the slickenside surface by root channels (RC). Position of roots is indicated by (R). (B) Line drawing of the soil clod in Fig. 2A.

differentiated features in the Ap horizon) to 100% (unidentified features in the Ap horizon). Coefficients of variation for areas contributed by root channels were 37% in the Ap horizon and 21% in the Bssk horizon; for ped interfaces they were 86% in the Ap horizon and 40% in the Bssk horizon. Hence, the variability is such that quantification of the impact of these morphological features on preferential flow would require large numbers of replicate observations.

Dye Movement in Pedons of Burleson Clay

Burleson Clay Microhighs

Undifferentiated features contributed to 96% of the stained area in the surface horizon of the Burleson microhigh (Table 4). Among other features transmitting dye in the surface horizon of the Burleson clay microhigh were ped interfaces and root channels.

In subsurface horizons of Burleson clay microhigh, the abundance of continuous root channels led to the preferential flow of dye solution along these features (37–78%). Coarse continuous roots created readily available macropore conduits, which were primarily responsible for transmitting the dye solution to a depth of 86 cm (Fig. 3). The contribution of ped interfaces was limited to less than 1% of the stained area. Slickenside sur-

TABLE 4

Percentages (as fraction of total stained area) of dye-stained features involved in solute transport at one site on Burleson clay microhigh and two sites on Burleson clay microlow

Horizon	Ped interfaces	Root channels	Undifferentiated features [†]	Unidentified features	Slickenside planes	Worm channels
Burleson Clay (Microhigh)						
A (0–20 cm)	2	2	96	<1	0	0
Bssk1 (20–63 cm)	<1	37	45	17	1	<1
Bssk2 (63–147 cm)	0	78	0	16	6	
Burleson Clay (Microlow)						
A (0–20 cm)						
Site 1	63	0	37	0	0	
Site 2	0	0	98	2	0	
BA (20–50)						
Site 1	86	1	0	13	0	
Site 2	30	6	45	19	0	
Bssk (50–166 cm)						
Site 1	0	0	0	100	0	
Site 2	94	2	0	4	1	
Depth of dye flow, cm						
Microhigh	86					
Microlow 1	69					
Microlow 2	67					

[†]Undifferentiated root channels and ped interfaces.

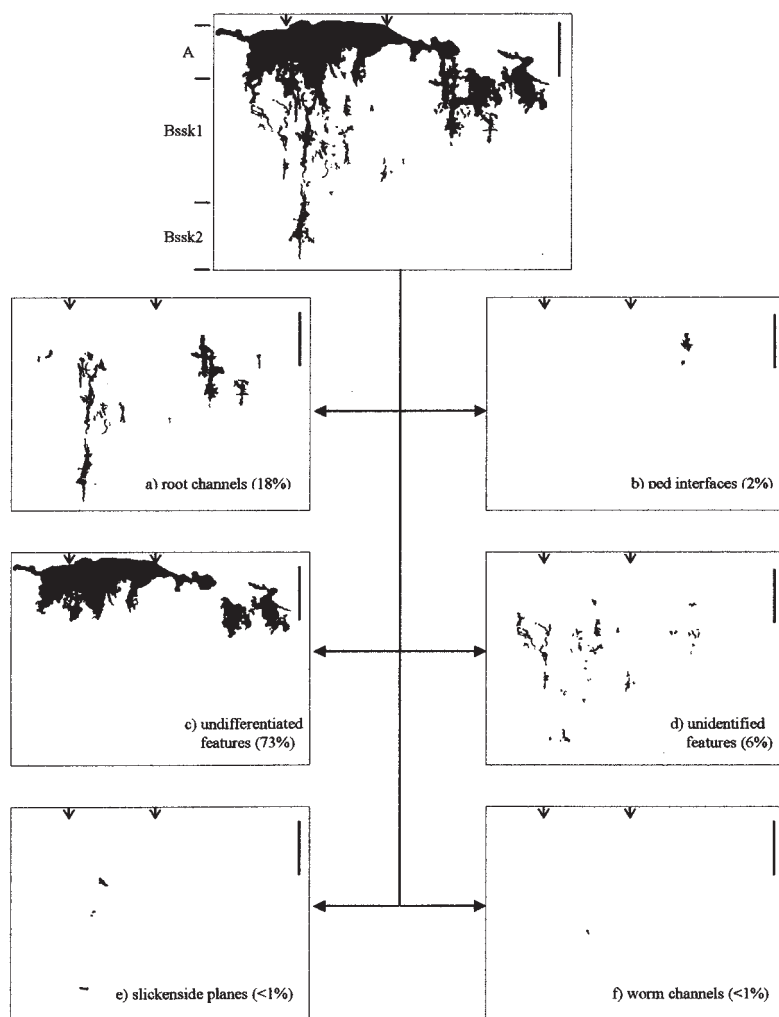


Fig. 3. Two-dimensional dye flow pattern in Burleson clay microhigh exposed along the central vertical line of intake zone (indicated by arrows). Overall dye-stained areas are separated into areas associated with (a) root channels, (b) ped interfaces, (c) undifferentiated features, (d) unidentified features, (e) slickenside planes, and (f) worm channels. The vertical bar indicates 20 cm.

faces contributed only about 1% of the stained area in the Bssk1 horizon and 6% in the Bssk2 horizon. In addition to the dye-stained slickenside faces detected during tracing of the dye pattern, slickenside planes stained with dye were observed later during sampling for thin sections. Dye transport along these features was conducted almost exclusively when the dye solution was transported to the surface of the slickenside by the adjacent root channel. The spatial distribution of dye in these cases did not seem to be limited to the confines of the root channel, with the dye solution moving along the slickenside surface. This was true when the root was positioned between slick-

enside planes as well as when very fine roots penetrated the surfaces of the slickensides.

A comparison was made between the extent of solute flow along the slickenside surfaces in Ships clay and Burleson clay. It was found that the areas of dye-stained slickenside surfaces increased from less than 1% in the Bssk horizon of Ships clay to 1% in the Bssk1 horizon and 6% in the Bssk2 horizons of the Burleson clay microhigh. However, it should be noted that dye flow along slickenside surfaces in Burleson clay was observed almost exclusively in one site of the the Burleson clay microhigh. Therefore, the lack of sites does not allow us to conclude whether the

increased involvement of slickensides in the transport of solutes in the subsoil Burleson clay microhigh was caused by the higher degree of slickenside planes development.

Burleson Clay Microlows

In the subsurface horizons of Burleson clay microlows, the solute flow was conducted predominantly by interpedal pores in both site 1 and site 2 (Table 4). Less than 1% of the stained area in the Bssk horizon of site 2 was stained by solute flow along slickenside surfaces (Fig. 4). However, this was the only case in the Burleson clay mi-

croflow where slickenside surfaces participated in water flow.

The Burleson clay microlows were characterized by greater solute flow along interpedal pores and less flow along root channels, compared with Burleson clay microhigh. It is possible that the decrease in the area stained by the dye transported along root channels was caused by the differences in vegetation between microhighs and microlows. The shallow root systems of vegetation in Burleson clay microlows (characterized by fine and very fine roots) did not create the readily available macropore conduits through which

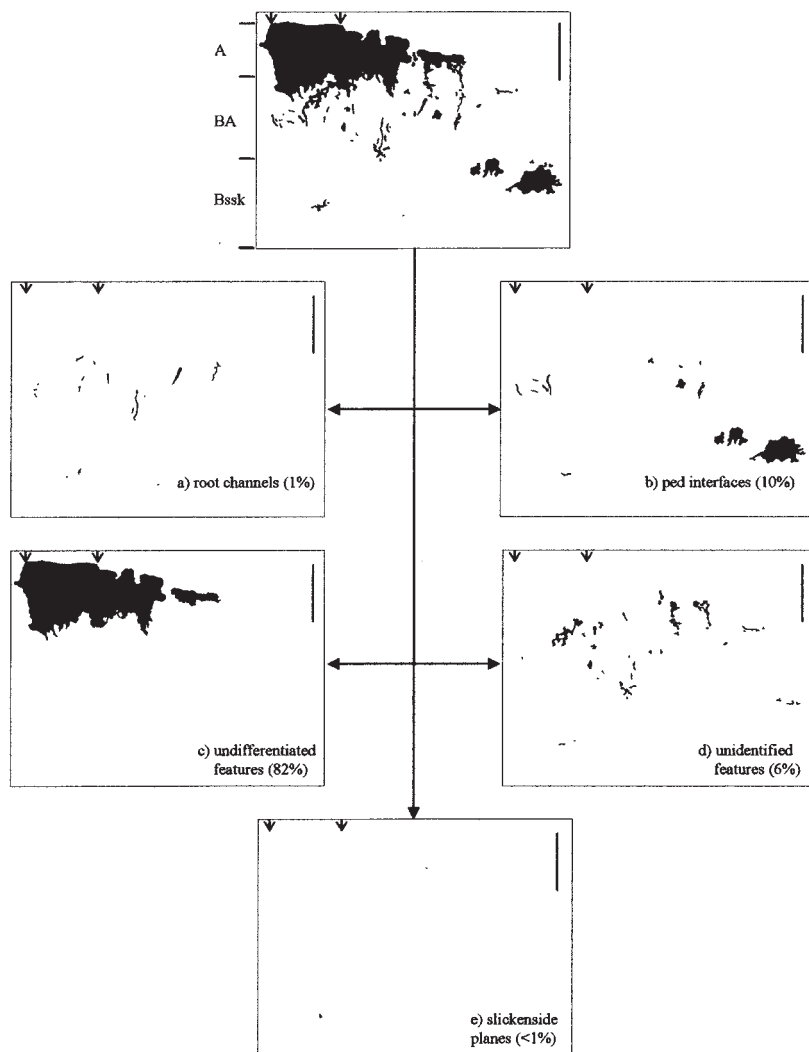


Fig. 4. Two-dimensional dye flow pattern in site 2 of Burleson clay microlow exposed along the central vertical line of intake zone (indicated by arrows). Overall dye-stained areas are separated into areas associated with (a) root channels, (b) ped interfaces, (c) undifferentiated features, (d) unidentified features, and (e) slickenside planes. The vertical bar indicates 20 cm.

the dye could be transported to greater depths, as in the pedon of Burleson clay microhigh. This conclusion is also supported by Mitchell et al. (1995), who found that in swelling clay soils, the taproot system of alfalfa produced the most stable macropores, compared with those produced by fine fibrous wheat roots and desiccation cracks. It is also likely that the fraction of interpedal pores conducting the dye in Burleson clay microlows would be transformed into cracks at dryer soil

conditions, through which the solute flow might be conducted.

Dye Movement in Pedons of Silawa Variant

Upon excavation of four sites of Silawa variant, it was observed that all stained features in the Ap and E horizons were primary packing void walls (walls of the voids caused by the random packing of the individual grains), further referred to as undifferentiated features (Table 5). A por-

TABLE 5

Percentages (as fraction of total stained area) of dye-stained features involved in solute transport at four sites on Silawa variant

Horizon	Ped interfaces	Root channels	Undifferentiated features [†]	Unidentified features	Termitarium
Ap+E (0–30 cm)					
Site 1	0	0	100	0	0
Site 2	0	<1	100	0	0
Site 3	0	0	100	0	0
Site 4	0	0	100	0	0
Mean			100		
SD			0		
CV			0		
Bt/E (0–30 cm)					
Site 1	<1	2	96	1	0
Site 2	9	6	84	1	0
Site 3	1	1	98	<1	0
Site 4	0	11	89	<1	0
Mean	3	5	92	1	
SD	4	4	6	0	
CV	132	79	6	0	
Bt1 (55–80 cm)					
Site 1	0	77	0	23	0
Site 2	62	38	0	0	0
Site 3	0	0	0	0	0
Site 4	0	69	0	0	31
Mean		46			
SD		30			
CV		60			
Bt2 (80–95 cm)					
Site 1	0	0	0	0	0
Site 2	0	0	0	0	0
Site 3	0	0	0	0	0
Site 4	0	100	0	0	0
Depth of dye flow, cm					
Site 1	76				
Site 2	85				
Site 3	50				
Site 4	110				
Mean	80				
SD	25				
CV	31				

[†]Undifferentiated root channels and packing voids.

tion of the dye became laterally distributed because of the slope gradient of about 1%. Traces of dye solution in the surface horizons were detected at a distance 1.5 m downslope from the dye intake zone.

The Bt/E horizon of Silawa variant was characterized by the presence of tongues and pockets of coarser surface soil material penetrating as tongues into the sandy clay loam material of the Bt1 horizon. A portion of the dye moved through these features in the Bt/E and Bt1 horizons of site 2 (Fig. 5). Solute flow along the tongue was not detected at a similar magnitude in the remaining three sites. Root channels, including those formed by very coarse (>10 mm) living pine roots, contributed to the majority of the solute flow in the subsurface horizons of all four sites. It was observed that in the horizons, Bt1 and Bt2 root channels served as primary pathways for transmitting the dye to depth. The transport of dye tracer along root channels was highly depen-

dent on the spatial variability of roots in the subsurface horizons. For example, the presence of a single continuous coarse root in site 4 resulted in the transport of dye solution to the depth of 110 cm. It served as only feature transporting the dye in horizons Bt1 (69%) and Bt2 (100%). Root channels, therefore, served as dominant pathways of solutes in three sites of Silawa variant, whereas in site 2, tongues of coarser infilling material also contributed to the solute flow.

Analysis of the data from four sites of Silawa variant has demonstrated high variability among groups of structural features transporting the dye in four sites of Silawa variant. For example, the coefficient of variation for stained areas contributed by undifferentiated features was 0% in the Ap and E horizons and 6% in the Bt/E horizons. The relatively low values of coefficient of variation for this group of features could be caused by the uniform dye staining along simple packing void walls. For the areas stained by dye

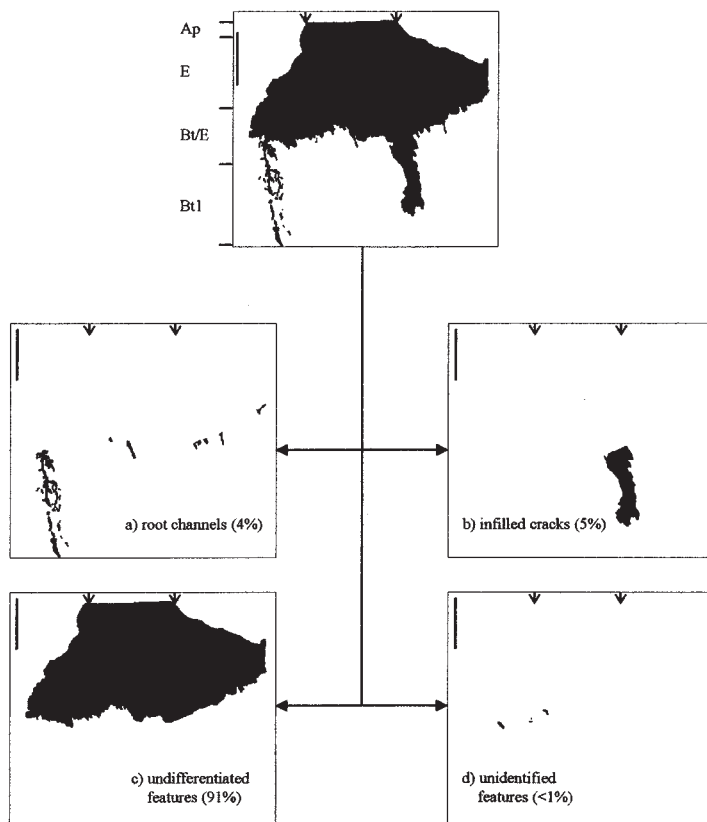


Fig. 5. Two-dimensional dye flow pattern in site 2 of Silawa variant exposed along the central vertical line of the intake zone (indicated by arrows). Overall dye-stained areas are separated into areas associated with (a) root channels, (b) infilled cracks, (c) undifferentiated features, and (d) unidentified features. The vertical bar indicates 20 cm.

moving along root channels, the coefficient of variation was 79% in the Bt/E horizon and 66% in the Bt1 horizon. In case of dye-stained ped interfaces, the coefficient of variation in the Bt/E horizon was 132%. The main reason for high variability in the latter two pathways of solute transport was the variance in spatial distribution of root channels and in-filled crack features in the subsoils of the four sites.

Quantification of Lateral Dye Movement in Relation to Vertical Intake Zone

In the four Ships clay sites, the percentage of stained features found outside the vertical intake zone varied from 20 to 39% (Table 6). In the profile of the Burleson clay microhigh, 81% of all stained features were located outside the vertical intake zone. In two sites of the Burleson clay microlow, 62 and 55% of stained area were found to be located outside the intake zone. In four sites of the Silawa variant, the percentages varied from 51 to 60%. It is thus apparent that although all the soils were located on nearly level to gently sloping geomorphic surfaces, lateral flow was still an important component of solute transport.

SUMMARY AND CONCLUSIONS

Effective flow area in the subsoil of Ships clay, Burleson clay microhigh, and Silawa variant at given moisture conditions was associated primarily with root channels. Ped interfaces served as primary pathways of solute transport. The involvement of various groups of void features in

the transport of water in the three studied structured soils was closely related to the type and abundance of vegetation, the continuity and interconnectivity of macropores such as root channels and interpedal pores, and macropore access to the infiltrating dye front. Future research on the comparison of pathways of solute flow in structured soils at various soil moisture tension values would be beneficial for developing a thorough understanding of water and solute flow distribution in these soils.

Slickenside planes seemed to play a minor role in the solute transport in both Vertisols, contrary to the commonly invoked concept. The limited involvement of slickenside features was likely caused by the lack of the macropore connectivity and the absence of readily available macropore pathways, such as open desiccation cracks, that would serve as means of solution delivery to the slickenside fissures. It is likely that further decrease in moisture content of Ships clay and Burleson clay would result in the increased involvement in solute transport of desiccation cracks and slickenside planes. The involvement of slickenside planes could also become more prominent with the increase in the amount of applied solutes during seasonal wet periods and high intensity rainfall events. Solute transport along slickenside planes was slightly higher in Burleson clay compared with Ships clay. However, the flow of solutes along these features was not prominent enough to determine whether the increased involvement was the result of either a higher degree of slickenside development in

TABLE 6
Lateral dispersion of the dye tracer solution outside vertical intake infiltration zone

	Total stained area, cm ²	Stained area outside vertical intake zone, cm ²	Total stained area outside vertical intake zone, %
<u>Ships clay</u>			
Site 1	536	190	36
Site 2	546	203	37
Site 3	891	176	20
Site 4	894	347	39
<u>Burleson Clay</u>			
Microhigh	1525	1235	81
Microlow 1	788	489	62
Microlow 2	1068	587	55
<u>Silawa variant</u>			
Site 1	2895	1467	51
Site 2	2828	1555	55
Site 3	3187	1817	57
Site 4	3268	1961	60

Burleson clay or a better-developed root system in Burleson clay microhighs.

The lateral transport of Brilliant Blue FCF dye was identified as one of the major components of the flow in the Alfisol and in both Vertisols even though the sites were located on nearly level to gently sloping geomorphic surfaces. The significant lateral transport in structural soils indicates the increase in the length of transport pathways and, consequently, the increase in the solute residence time to depth. Because a considerable amount of solutes in these soils was transported laterally, caution needs to be exercised when calculating the hydraulic conductivity of these soils on the basis of cross-sectional area.

REFERENCES

- Beck, A. J., and K. C. Jones. 1996. The effects of particle size, organic matter content, crop residues and dissolved organic matter on the sorption kinetics of atrazine and isoproturon by clay soil. *Chemosphere* 32:2345–2358.
- Bouma, J., and L. W. Dekker. 1978. A case study on infiltration into dry clay soil. I. Morphological observations. *Geoderma* 20:27–40.
- Clothier, B. E., and S. R. Green. 1997. Roots: The big movers of water and chemicals in soil. *Soil Sci.* 162:534–543.
- Cunningham, S. D., T. A. Anderson, A. P. Schwab, and F. C. Hsu. 1996. Phytoremediation of soils contaminated with organic pollutants. *Adv. Agron.* 56:55–114.
- Flury, M., and H. Flühler. 1994. Brilliant Blue FCF as a dye tracer for solute transport studies – A toxicological overview. *J. Environ. Qual.* 23:1108–1112.
- Flury, M., and H. Flühler. 1995. Tracer characteristics of Brilliant Blue FCF. *Soil Sci. Soc. Am. J.* 59:22–27.
- Gish, T. J., D. Gimenes, and W. J. Rawls. 1998. Impact of roots on ground water quality. *Plant Soil* 200:47–54.
- Gish, T. J., and W. A. Jury. 1983. Effect of plant roots and root channels on solute transport. *Trans. ASAE* 26:440–444.
- Kanwar, R. S., and A. Bakhsh. 2001. Preferential flow and its effects on nitrate and herbicide leaching to groundwater under various tillage systems. *In* Preferential Flow: Water Movement and Chemical Transport in the Environment. Proc. 2nd Int. Symp., Honolulu, Hawaii, January 3–5, 2001. ASAE, St. Joseph, MI, pp. 165–168.
- Kissel, D. E., J. T. Ritchie, and E. Burnett. 1973. Chloride movement in undisturbed swelling clay soil. *Soil Sci. Soc. Am. Proc.* 37:21–24.
- Kutilek, M. 1996. Water relations and water management of Vertisols. *In* Vertisols and Technologies for Their Management. N. Ahmad and A. Mermut (eds.). Elsevier Science Publishers, Amsterdam, pp. 200–230.
- Lawes, J. B., J. H. Gilbert and R. Warington. 1882. On the amount and composition of the rain and drainage water collected at Rothamstead. Williams, Clowes and Sons, Ltd., London.
- Lin, H. S., K. J. McInnes, L. P. Wilding, and C. T. Hallmark. 1996. Effective porosity and flow rate with infiltration at low tensions into a well-structured subsoil. *Trans. ASAE* 39:131–135.
- Mitchell, A. R., T. R. Elsworth, and B. D. Meek. 1995. Effect of root systems on preferential flow in swelling soils. *Commun. Soil Sci. Plant Anal.* 26 (15&16):2655–2666.
- Mowery, I. C., H. Oakes, J. D. Rourke, F. Matanzo, and H. L. Hill. 1958. Soil survey of Brazos County, Texas. USDA Soil Conservation Service. U.S. Govt. Printing Office, Washington, DC.
- Nobles, M. M., L. P. Wilding, and K. J. McInnes. 2003. Soil structural interfaces in some Texas Vertisols and their impact on solute transport. *In* Achievements in Micromorphology. G. Stoops (ed.). *Catena* 54/3:477–493.
- Petersen, C. T., S. Hansen, and H. E. Jensen. 1997. Tillage-induced horizontal periodicity of preferential flow in the root zone. *Soil Sci. Soc. Am. J.* 61:586–594.
- Soil Survey Staff. 1996. Soil Survey Laboratory Methods Manual. Soil Survey Investigations Report No. 42. USDA Natural Resources Conservation Service. U.S. Govt. Printing Office, Washington, DC.
- Stamm, C., H. Flühler, R. Gächter, J. Leuenberger, and H. Wunderli. 1998. Preferential transport of phosphorous in drained grassland soils. *J. Environ. Qual.* 27:515–522.
- Steenhuis, T. S., and J. Y. Parlange. 1991. Preferential flow in structured and sandy soils. *In* Preferential Flow. T. J. Gish and A. Shirmohammadi (eds.). Proc. Nation. Symp., Chicago, IL. ASAE, St. Joseph, MI, pp. 12–21.
- Turner, R. R., and K. F. Steele. 1988. Cadmium and manganese sorption by soil macropore linings and fillings. *Soil Sci.* 145:79–86.
- Wilding, L. P., and D. Tessier. 1988. Genesis of Vertisols: Shrink-swell phenomena. *In* Vertisols: Their distribution, Properties, Classification and Management. L. P. Wilding and R. Puentes (eds.). Texas A&M Univ. Printing Center, College Station, TX, pp. 55–82.